The composition and architecture of membranes

Structural lipids in membranes

- Plasma membranes act as intelligent barriers that isolate cells, in such a way that it is the cell itself that decides which extracellular signals can reach the interior of it

- Likewise, the organelle membranes allow the cell to be compartmentalized, in such a way that substances can accumulate inside it, which are released in a controlled way to the cytoplasm, or separate metabolic processes.

- The structural characteristic of membranes is that they are formed by a lipid bilayer.

- In order to understand the function of membranes, it is necessary to know their molecular composition: polar lipids, proteins and carbohydrates bound to proteins and lipids.

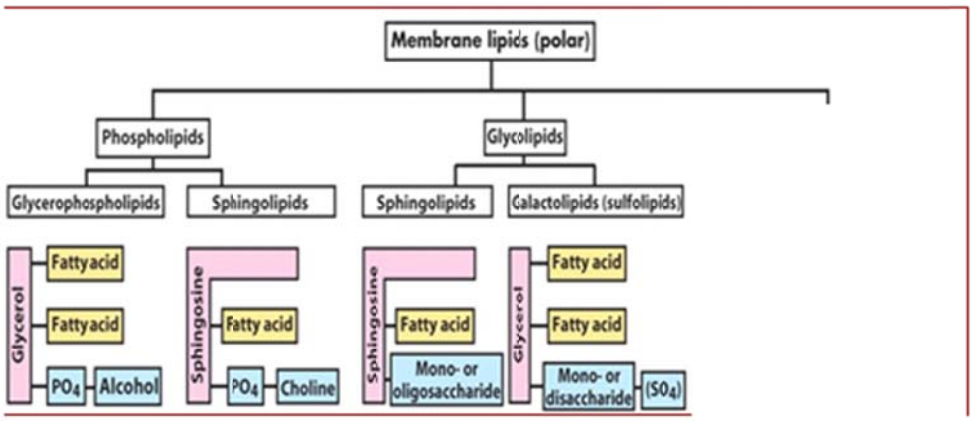
- Common to all membrane lipids is that they are amphipathic molecules, that is, one end is hydrophilic and the other hydrophobic.

- The different membrane lipids are

    - Lipids with phosphate: Glycerophospholipids and Sphingomyelins

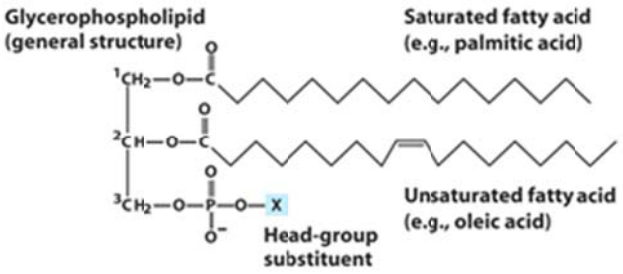
    - Lipids with sugars glucosphingolipids, cerebrosides and gangliosides

    - Cholesterol

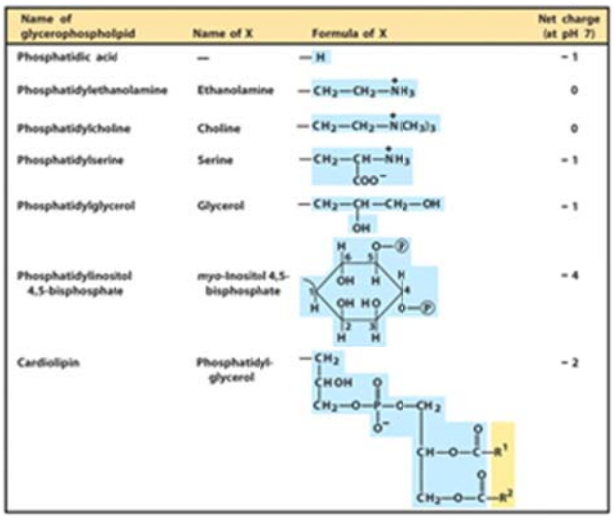


Glycerophospholipids

- They are formed by glycerol 3 phosphate molecule esterified with two fatty acids at carbons 1 and 2



- Generally, the fatty acid that esterifies carbon 1 is saturated with 16 or 18 carbons, while that which esterifies at carbon 2 is unsaturated at 18 or 20 C



- They are differentiated by an alcohol that esterifies the carbon 3 phosphate: choline, ethanlolamine, serine, inositol

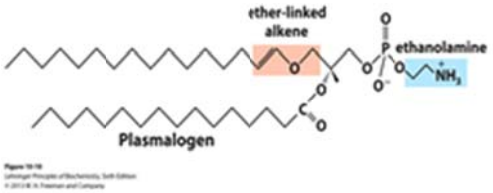
- Phosphatidylcholine, phosphatidylethanolamine and phosphatidylserine are the most abundant in plasma membranes

- Phosphatidylinositol is a phospholipid with a very important role in the transduction of signals from the membrane to the interior; Inositol can be found phosphorylated on several of the –OH groups and can act as a recruiter for

signaling molecules and can be released from the membrane as a second messenger.

- Cardiolipins are preferentially located in the inner membrane of the mitochondria

- In the membranes of heart tissue there is a special type of glycerophospholipid in which the ether function is present.

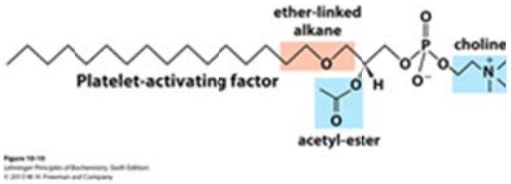


- In these lipids one of the fatty acids is substituted by an alkyl or alkenyl hydrocarbon chain.

- One of the lipids with the most abundant ether function are **plasmalogens**; they are characterized by having an unsaturated chain that is linked by an ether bound to carbon 1, and phosphocholine as the head group.

- They are very abundant in the mitochondrial membranes, especially in the heart.

- Another lipid with an ether function is the **platelet activating factor**, a molecule secreted by white blood cells that stimulates platelet aggregation and the release of serotonin inducing vasoconstriction.



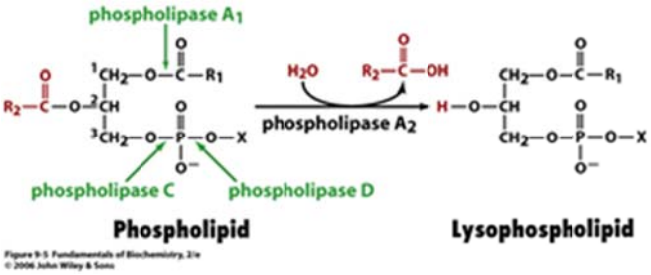
Glycerophospholipids as signal transducers

- Phospholipids can be split into their different components (fatty acids, phosphate, head alcohol) through the activity of enzymes called phospholipases

- There are three types of phospholipases:

  - PLPA1: hydrolyze and release the fatty acid attached to C1

  - PLPA2: hydrolyze and release the fatty acid attached to C2



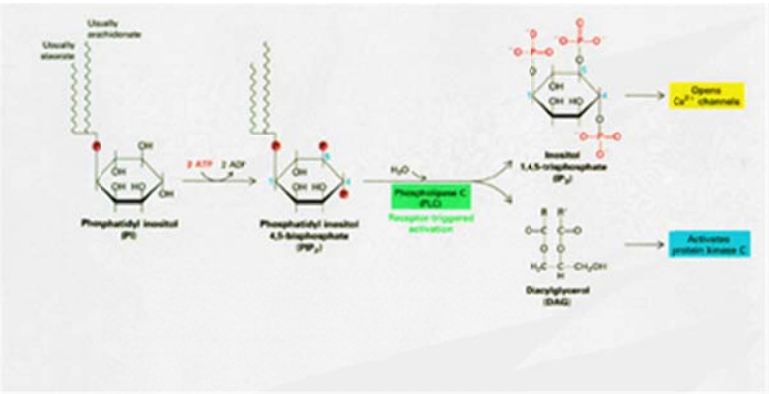
  - PLPC: hydrolize the phosphoester bond that binds the phosphate with the C3 releasing the head's phosphogroup

  - PLPD: hydrolyze and release the head alcohol

- The activity of these enzymes is closely regulated, and the products that derive from their hydrolysis are frequently molecules with biological activities or precursors of molecules with biological activities (ie arachidonic acid is precursor of eicosanoids)

- As an example, membrane phosphatidylinositols are precursors of two molecules with important biological activities: IP3 and DAG

- The hydrolysis mediated by PLPC, activated as a consequence of an extracellular signal, on phosphatidyl inositol 4,5-bisphosphate releases two second messengers: diacylglycerol (DAG) and inositol 1,4,5-trisphosphate (IP3)

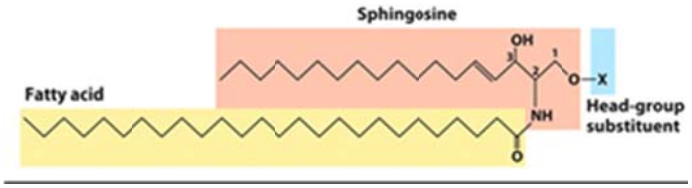


- IP3 is soluble and, through its receptors in the RER, causes the release of Ca^{++}

- This Ca^{++} along with the DAG activate protein kinase C (PKC), which phosphorylates several proteins causing conformational changes that affect their activity

Sphingolipids

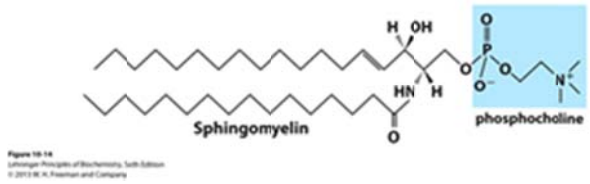
- They also have a polar head and two nonpolar tails like glycerophospholipids, but unlike these, they do not contain glycerol.



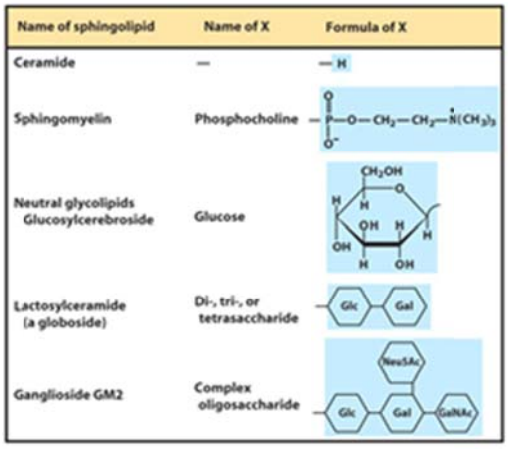
- They have fatty acid but now it is attached to an 18-carbon, unsaturated amino alcohol called sphingosine

- The simplest sphingolipid is ceramide, the rest of the sphingolipids contain different head groups attached to carbon 1 of sphingosine:

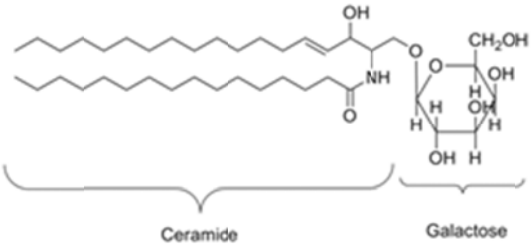
- **sphingomyelins** have phosphocholine or phosphoethanolamine, they look a lot like glycerophospholipids; are mostly present in plasma membranes and in the myelin sheaths of axons



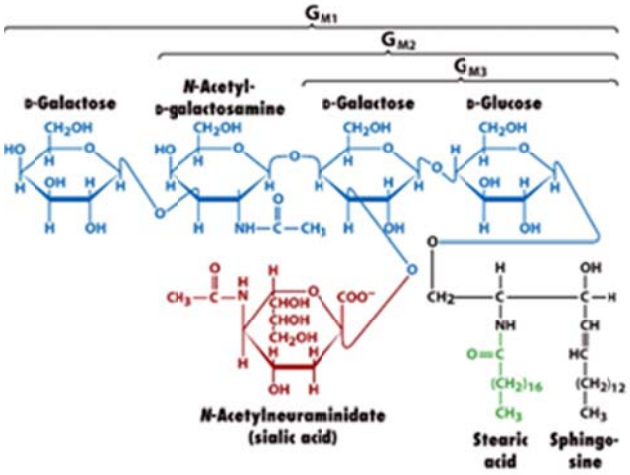
- **glucosphingolipids**, have sugars as head group and are mainly located on the extracellular side of the membranes. One of their most important functions is to act as recognition signals between cells and receptor molecules. There are different types depending on the type and number of sugar units at the polar head:



- **cerebrosides:** they have a monosaccharide attached to the ceramide; galactocerebrosides are very abundant in nervous tissue, while glucocerebrosides are frequent in other tissues

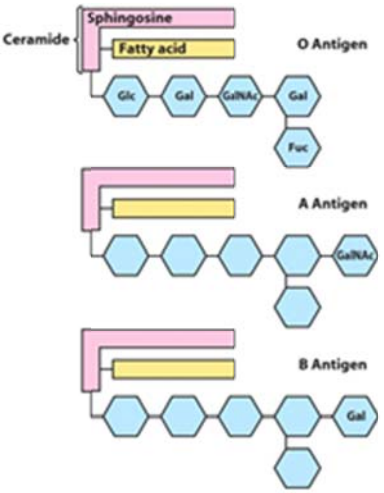


- **globosides**: they have two or more sugars attached to the ceramide, but none of them have an electrical charge.
- **gangliosides**: they have oligosaccharides branched with sialic acid residues, so they have a negative charge at pH 7.



- Sphingolipids, which are mostly located in the outer half-layer of the plasma membrane, have very clear functions in cell-cell recognition processes.

- Blood groups ABO are determined by the presence in cells of different types of glycosphingolipids

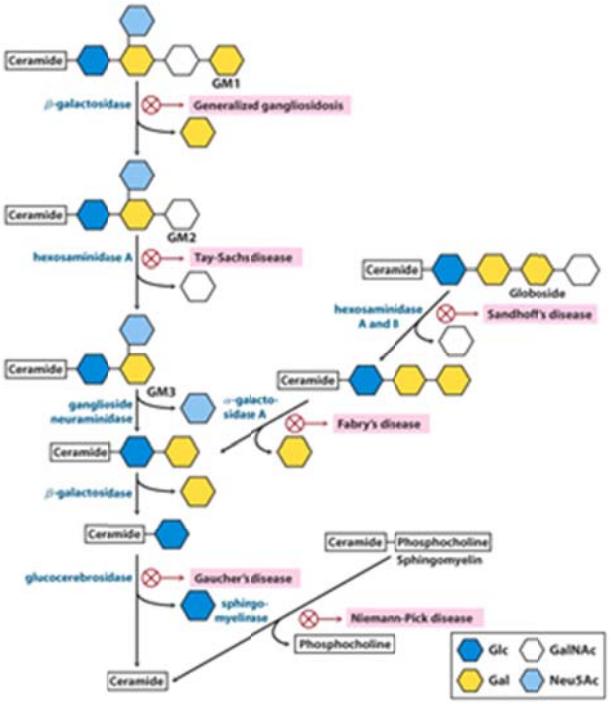


Abnormal accumulations of membrane lipids

- The polar lipids of membranes undergo constant metabolic turnover, the rate of their synthesis normally counterbalanced by the rate of breakdown.

- The breakdawn takes place in the lysosomes where the enzymes involved are stored

- For each of the bonds that join the different molecules part of sphingolipids there is an specific enzyme.



- The degradation of gangliosides is especially complex due to the large number of enzymes involved in the hydrolysis of chemical bonds

- When the degradation of sphingolipids is altered by defects of any of the enzymatic activities, their accumulation in the cell causes serious consequences, especially at the neurological level.

- For example, Niemann-Pick disease is caused by a deficiency of the enzyme sphingomyelinase. This causes sphingomyelin to accumulate in the brain, spleen, and liver, causing delayed neurological development and very early death.

Sterols

- They are made up of four fused rings that constitute the steroid nucleus.



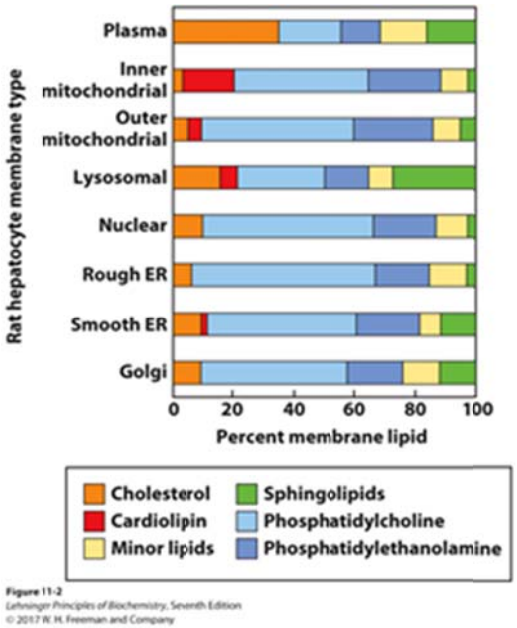
- The main sterol is cholesterol; is an amphipathic molecule due to the presence of an -OH at C3 and a hydrocarbonated side chain attached to C17, which is almost as long as a fatty acid

- Cholesterol is an important factor in the fluidity and permeability of the membrane since it occupies the gaps left by other molecules. The higher the amount of cholesterol, the less permeable and fluid the membrane is.

- In addition to its role in membranes, it is a precursor of other products with biological activity: steroid hormones, bile salts and vitamin D

Each type of membrane has a characteristic lipid and protein composition

- The proportions of lipids and proteins are different in each type of membrane, which causes them to have different biological roles.



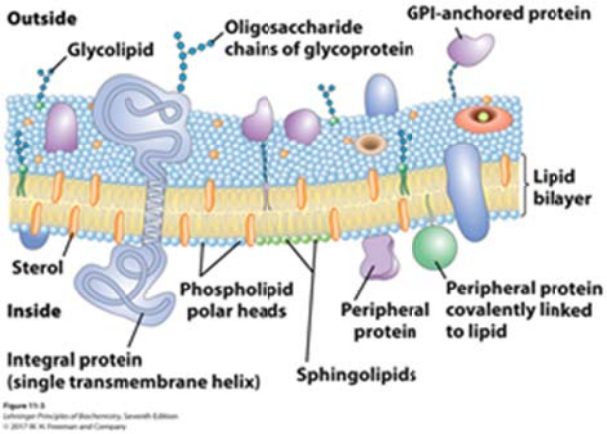
- Some types of neurons have their axons covered by myelin sheaths that insulate them.

- On the other hand, the internal mitochondrial membranes contain a high proportion of proteins that constitute the respiratory complexes that allow the transport of electrons and the consequent synthesis of ATP

- There are also big differences regarding the lipid composition. Plasma membranes are high in cholesterol and sphingolipids

- Mitochondrial membranes have very little cholesterol and sphingolipids but large amounts of glycerophospholipids and in the case of the inner membrane, cardiolipins.

- The composition of proteins also varies, even more widely than lipids, reflecting the functional specialization of the membrane.



- It is also very common for membrane proteins to be covalently linked to oligosaccharides through O-glucosidic bonds with Ser and Thr, or N-glucosidic bonds with Asn.

- These oligosaccharides are essential for the proper functioning of proteins:

  - necessary for its correct folding and stability

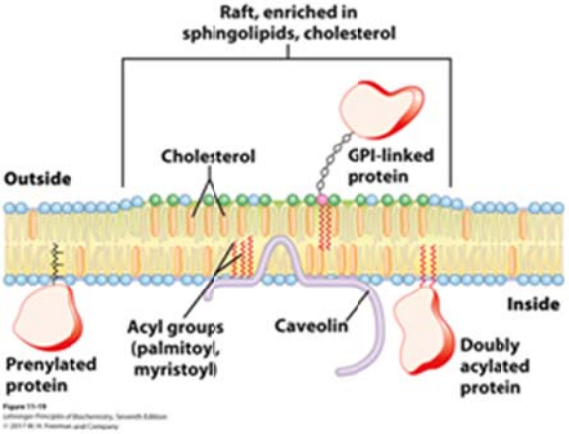
  - necessary for the correct localization of the protein in the appropriate membrane

  - in some cases they are necessary as sites of interaction of ligands and receptors

Lipid rafts in the membranes

- There are regions of the membrane (especially the plasmatic) that are rich in sphingolipids and cholesterol, which makes the bilayer thicker and less fluid than in neighboring areas.

- These regions (microdomains) are called lipid rafts.



- It seems that they can act as recruitment points for recipients and that they can facilitate interaction between them

- They also have a relevant role in endocytosis processes, such as receptor-mediated endocytosis.

- In this process, the caveolin protein is associated with the internal half-layer of the plasma membrane, causing its curvature and forming the caveola

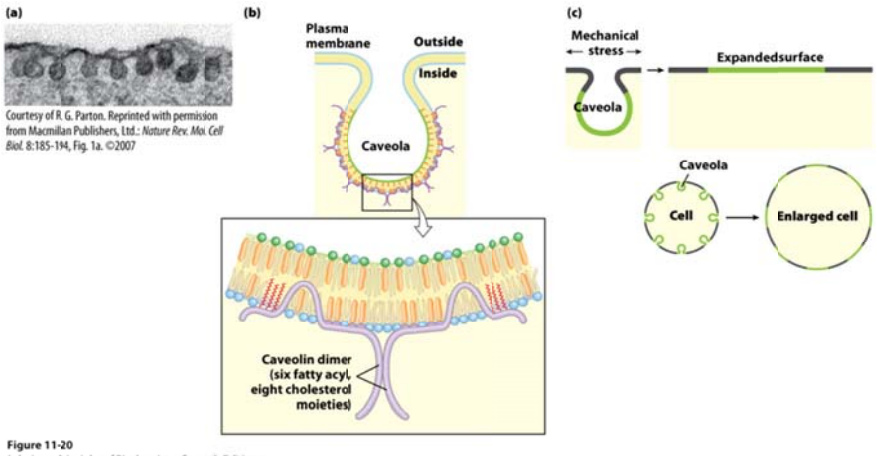


Figure 11-20
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